Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PCH805S – ORGANIC CHEMISTRY - II

Time : 3 hrs **Max Marks :75**

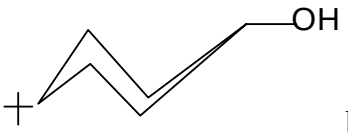
SECTION – A (20X1=20)

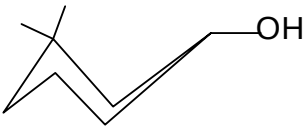
Answer **ALL** Questions

I. Choose the correct answer

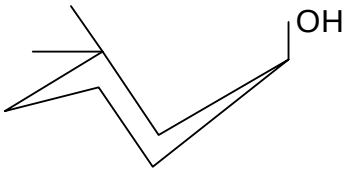
1. The stable conformation of 1,2-dichloroethane is
   a) Anti b) Gauche c) Meso d) Staggered

2. Which of the following undergoes oxidation rapidly?

a) 

b) 

c)

d) 

3. Order of reactivity is SN^1reaction
   a) Tert.halides>prim.halides>sec.halides>CH3X
   b) Tert.halides>CH_3X>sec.halides>prim.halides
   c) Tert.halides>sec.halides>prim.halides>CH_3X
   d) Prim.halides>sec.halides>sec.halides>tert.halides>CH_3X

4. The reactivity order of alkyl groups in SE^2 reaction is
   a) Me>ET>propyl>oso-propyl>neopentyl
   b) Me>ET>propyl>neo-pentyl>iso-propyl
   c) Me>ET>neopentyl>Propyl>isopropyl
   d) ET>me>neopentyl>propyl>isopropyl

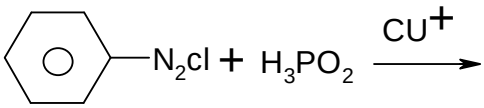
5. The product formed is Dieckmann condemnation is
   a) Cyclic ketones b) Acylic ketones
   c) Diesters d) Dibasic acids

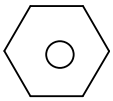
6. Neomenthyl xanthate during chugaev reaction yields major-product as
   a) Menth-3-ene b) Menth-2-ene
   c) Menth-2-one d) Menth-3-one

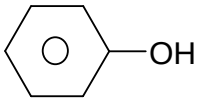
7. The electrophile in nitration reaction of benzene is
   a) NO+ b) NO_2^+ c) HNO_2 d) NO_3^-

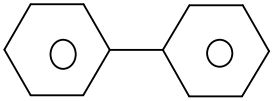
8. Pyrrole on nitration gives
   a) 2-nitropyrrole b) 3-nitropyrrole
   c) 4-nitropyrrole d) 5- nitropyrrole

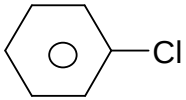
9. Chichibabin reaction is
   a) Nucleophilic aromatic substitution
   b) Nucleophilic aliphatic substitution
   c) Electrophilic aromatic substitution
   d) Electropilic aliphatic substitution

10. 

a) 

b) 

c) 

d) 

II. Fill up the blanks

11. In cis-decalin, both cyclohexane rings are joined by_____bonds.

12. Rate of acetolysis is of antitosylates is faster than its syn isomer due to_____.

13. Stobbe reaction is limited to those esters which do not undergo _____ condemnation.

14. In vilsmeyer reaction, the formylating agent is _____.

15. Benzyne intermediate is formed by _____ mechanism.

III. Match the following

16. Cis-9-methyl decalin	a) Intramolecular
17. Enamine	b) Formylation
18. Dieckmann condensation	c) Amination
19. Riemer-Tiemann reaction	d) Nucleophile
20. Chichibabin reaction	e) Hydrolysis
	f) Racemic mixture

SECTION – B (5X4=20)

Answer any **FIVE** Questions

21. Draw the Newman projection of trans-decalin.

22. What are the two factors that decide conformationaltability.

23. What is Hell-Volhard-Zelinsky reaction?

24. "Allylic substrates undergo SN^1 reaction rapidly" Explain this fact.

25. What is neighbouring group participation?

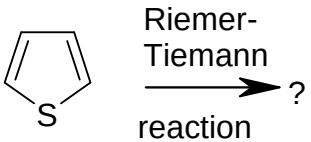
26. What is Aldol condensation?

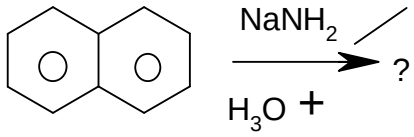
27. Define saytzeff's rule.

28. Give the mechanism of cope elimination.

29. Give any two electrophilic substitutions of pyrrole.

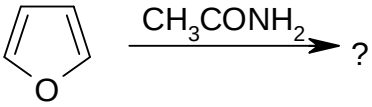
30. Complete the following

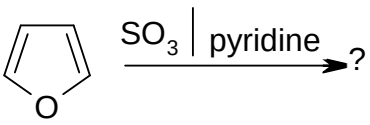
a) 

b) 

31. Give any one method for the generation of benzyne intermediate.

32. Complete the following

a) 

b) 

SECTION- C (5X7=35)

Answer **ALL** Questions

33. a) "Cis – 1,2-dimethyl cyclohexane is non-resolvable though it is disymmetric"-explain this fact. **(4)**
    b) Explain the observation that cis-decalin is a non-resolvable racemate. **(3)**

34. a) Discuss the conformation and saponification of ethyl-4-t-butylcyclohexane carboxylate.

35. Discuss the mechanism and stereo chemistry of SE^i reactions.

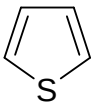
36. Predict the products and explain

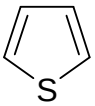
a) $CH_3\text{-}CH_2\text{-}Br \xrightarrow{C_2H_5O^-} ?$

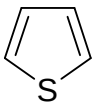
b) $H_3C\text{-}CH_2\text{-}C(OH)(CH_3)\text{-}CH_3 \xrightarrow[H_2O]{H^+} ?$

37. Give the mechanisms of the following condensation.
   i) Stobbe condensatiion
   ii) Perkin condensation

38. a) Give the mechanism of Gattermann reaction. **(4)**

b) i)  $\xrightarrow[AC_2O]{HNO_3}$?

ii)  $\xrightarrow[Hcl]{HCHO}$?

iii)  $\xrightarrow{I_2/HgO}$?

39. a) Write a note on Ziegler alkylation. **(3)**
   b) Write any four nucleophilic substitution reaction involving diazonium ions.

****